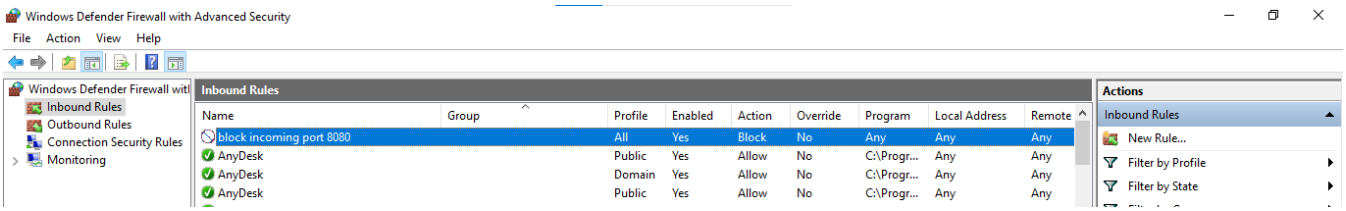
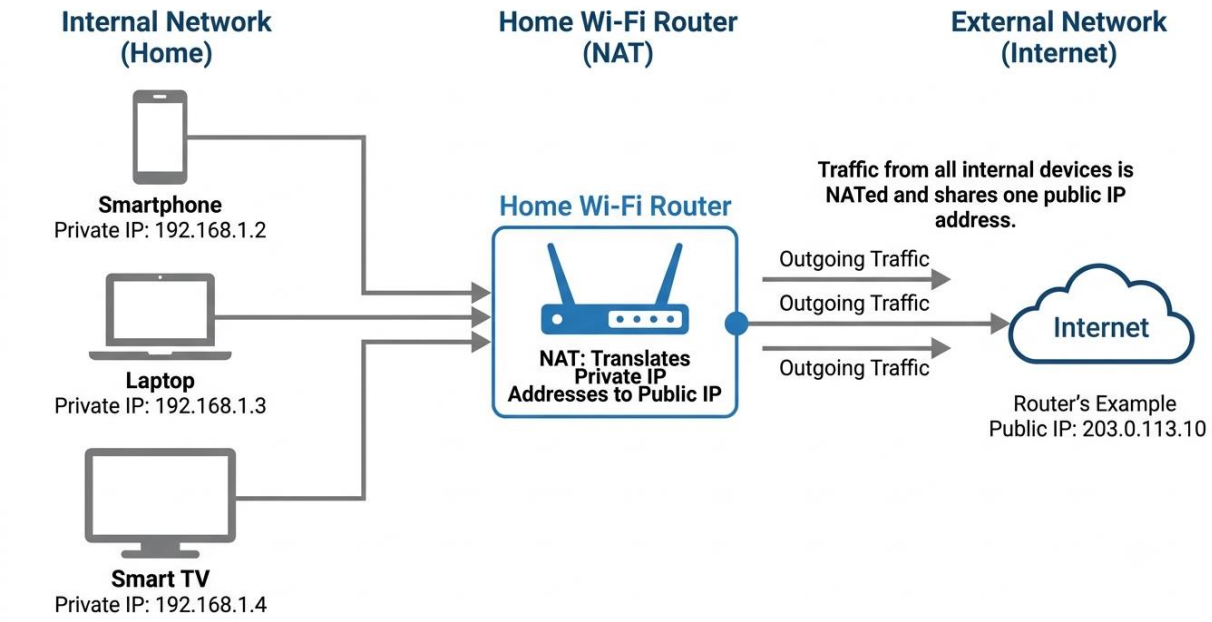
Secure Networks with Firewalls, NAT, Port Filtering, Packet Filtering, and Other Methods

1. Configure Windows Defender Firewall



2. Simulate NAT (Network Address Translation)



Working:

NAT (Network Address Translation) is a technique used by a router to translate private IP addresses into a public IP address for internet access. It allows multiple devices, such as a smartphone, laptop, and smart TV, to share one public IP address when accessing the internet.

3. Configure Port Filtering

Steps:

1. Open a web browser and enter the Wi-Fi router's IP address.
2. Log in to the router administration panel.
3. Open the Firewall, Security, or Access Control settings.
4. Find the Port Filtering or Service Filtering option.
5. Create a new rule to block outgoing TCP traffic on port 25 (SMTP).
6. Set the direction to Outbound and the action to Block or Deny.
7. Save and apply the settings.

Note: The router used for this assignment does not provide access to port filtering settings, so the rule could not be configured physically.

How does blocking port 25 help prevent spam?

Blocking outgoing port 25 can prevent devices on the network from directly sending email through standard SMTP connections. It can reduce spam from infected devices, although it does not block every method of sending spam because other ports and email services may still be used.

4. Packet Filtering Rules

Packet filtering checks packet information such as source IP address, destination IP address, protocol, and port number to allow or block network traffic.

Sample Packet Filtering Rules:

1. Allow HTTP traffic on TCP port 80.
2. Allow SSH traffic on TCP port 22 only from authorized IP addresses.
3. Allow DNS traffic on UDP port 53.

Packet Analysis:

No.	Packet Details	Decision	Reason
1	192.168.1.5 → 172.217.166.78:80 (HTTP)	Allowed	TCP port 80 is allowed.
2	10.0.0.10 → 34.201.23.1:22 (SSH)	Blocked	The source IP is not specified as an authorized IP.
3	192.168.1.7 → 8.8.8.8:53 (DNS)	Allowed	UDP port 53 is allowed.

Conclusion:

Packet filtering helps control network traffic by checking IP addresses, protocols, and port numbers. It improves network

security by allowing trusted traffic and blocking unauthorized connections.

Note: These decisions are based on the sample rules above because the trainer's actual rule set was not provided.

5. Uses of Firewalls and Packet Filtering by Instagram and Zomato

1. Protection Against Unauthorized Access

Instagram and Zomato may use firewalls to block unauthorized connections and allow only permitted network traffic to their servers.

2. Protection Against Cyberattacks

They may use packet filtering and firewalls to identify and block suspicious traffic, helping protect their servers against certain network attacks.

3. Port Filtering and Traffic Control

They may restrict access to unnecessary ports and allow only required services. This reduces the attack surface and helps prevent unauthorized access to server services.

Conclusion

Firewalls, NAT, port filtering, and packet filtering help improve network security. These methods control network traffic, reduce unauthorized access, and protect systems from various cyber threats.